

Bayesian Computational Statistics

Bayes' Theorem, Rejection Sampling,
Importance Sampling and Importance Resampling

Ravleen Bajaj

1 Bayes' Theorem and Bayesian Inference

Let θ denote the parameters of interest and y denote the observed data.

Bayes' Theorem [1, Ch. 1] combines the prior distribution $p(\theta)$ with the sampling model (likelihood) $p(y | \theta)$ to obtain the posterior distribution $p(\theta | y)$:

$$p(\theta | y) = \frac{p(y | \theta) p(\theta)}{p(y)} = \frac{p(y | \theta) p(\theta)}{\int p(y | \theta) p(\theta) d\theta} \quad (1)$$

Where:

- $p(\theta)$ is the **prior distribution**, representing belief before seeing the data.
- $p(y | \theta)$ is the **likelihood function** when viewed as a function of θ .
- $p(\theta | y)$ is the **posterior distribution**.
- $p(y) = \int p(y | \theta) p(\theta) d\theta$ is the **marginal likelihood** (or evidence), which acts as a normalizing constant.

In practice, $p(y)$ is often intractable, so we work with the unnormalized posterior density:

$$p(\theta | y) \propto q(\theta | y) = p(y | \theta) p(\theta) \quad (2)$$

2 Direct Simulation and Rejection Sampling (Section 10.3)

When independent samples cannot be drawn directly from $p(\theta | y)$, we use proposal distributions [1, Sec. 10.3]. Robert and Casella call g the **instrumental** (or candidate) density and f the **target** density [2, Sec. 2.3].

2.1 Rejection Sampling

To draw samples from an unnormalized target density $f(\theta) \propto p(\theta | y)$, choose a proposal distribution $g(\theta)$ such that for some known constant $M > 0$:

$$f(\theta) \leq M g(\theta) \quad \text{for all } \theta \text{ where } f(\theta) > 0. \quad (3)$$

Algorithm 1 Rejection Sampling

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1: repeat
2:   Sample a candidate  $\theta^* \sim g(\theta)$ .
3:   Sample  $U \sim \text{Uniform}(0, 1)$ .
4:   Calculate acceptance ratio  $r = \frac{f(\theta^*)}{M g(\theta^*)}$ .
5: until  $U \leq r$ 
6: return  $\theta = \theta^*$  as a draw from  $p(\theta | y)$ .
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Properties and limitations.

- The accepted points θ are independent draws from the exact target distribution.
- If f and g are both normalized densities, the acceptance probability is $1/M$ and the number of candidates needed per accepted draw is geometric with mean M [2, Sec. 2.3]. If f is only known up to a constant, the acceptance probability is $\int f(\theta) d\theta / M$.
- Only the functional form of f is needed, up to a multiplicative constant, but the bound M must be known.
- Finding an efficient M becomes exponentially difficult in high dimensions.

Why it works: the fundamental theorem of simulation.

Simulating $\theta \sim f$ (normalized) is equivalent to simulating (θ, U) uniformly on the region under the graph of f ,

$$\{(\theta, u) : 0 < u < f(\theta)\},$$

and keeping the θ -coordinate [2, Sec. 2.3]. Accept–Reject samples uniformly under the envelope $M g(\theta)$ (draw $\theta^* \sim g$, $u = U M g(\theta^*)$) and keeps the points that fall under f , i.e. those with $U \leq f(\theta^*)/(M g(\theta^*))$.

3 Importance Ratio and Importance Sampling (Section 10.4)

Importance sampling [1, Sec. 10.4] is a variance-reduction Monte Carlo method used to compute expectations of functions $h(\theta)$ with respect to the target distribution $p(\theta | y)$ without requiring exact draws.

3.1 Importance Ratio

If target distribution $p(\theta | y) \propto q(\theta | y)$ and proposal distribution $g(\theta)$ are used, the **importance ratio** (or raw importance weight) for a point θ^s is:

$$w(\theta^s) = \frac{p(\theta^s | y)}{g(\theta^s)} \quad \text{or, using unnormalized densities,} \quad w(\theta^s) = \frac{q(\theta^s | y)}{g(\theta^s)} \quad (4)$$

3.2 Importance Sampling Expectation Estimator

To estimate $\mathbb{E}[h(\theta) \mid y] = \int h(\theta)p(\theta \mid y)d\theta$:

Draw S independent samples $\theta^1, \theta^2, \dots, \theta^S \sim g(\theta)$ and calculate normalized weights:

$$\tilde{w}^s = \frac{w(\theta^s)}{\sum_{j=1}^S w(\theta^j)} \quad (5)$$

The importance sampling estimator is given by:

$$\hat{\mathbb{E}}[h(\theta) \mid y] = \sum_{s=1}^S \tilde{w}^s h(\theta^s) = \frac{\sum_{s=1}^S w(\theta^s) h(\theta^s)}{\sum_{s=1}^S w(\theta^s)} \quad (6)$$

Key considerations.

- Requires $g(\theta) > 0$ whenever $p(\theta \mid y) > 0$.
- The tail behavior of $g(\theta)$ should be heavier than $p(\theta \mid y)$; otherwise, $w(\theta)$ can have unbounded variance.

3.3 Properties of the Estimator

Importance sampling [2, Sec. 3.3] could be represented through the identity

$$\mathbb{E}_f[h(\theta)] = \int h(\theta) \frac{f(\theta)}{g(\theta)} g(\theta) d\theta = \mathbb{E}_g \left[h(\theta) \frac{f(\theta)}{g(\theta)} \right], \quad (7)$$

valid whenever the support of g contains the support of f . For normalized f and g , the estimator

$$\hat{\mu}_{\text{IS}} = \frac{1}{S} \sum_{s=1}^S h(\theta^s) \frac{f(\theta^s)}{g(\theta^s)}, \quad \theta^s \sim g, \quad (8)$$

is unbiased and converges almost surely to $\mathbb{E}_f[h(\theta)]$ by the strong law of large numbers.

Variance and the choice of g .

- The variance of $\hat{\mu}_{\text{IS}}$ is finite only if $\mathbb{E}_g[h^2(\theta) f^2(\theta)/g^2(\theta)] < \infty$.
- If f/g is unbounded, which happens when g has lighter tails than f , the variance is typically infinite and the estimator is unreliable.
- The variance-minimizing instrumental density for a fixed h is proportional to $|h(\theta)| f(\theta)$. It is rarely usable directly, because its normalizing constant is $\int |h|f$, but it suggests putting mass where $|h|f$ is large.
- Since g need not equal f , importance sampling can have *lower* variance than averaging h over draws from f itself.

Self-normalized estimator.

When f or g is known only up to a constant (as with the unnormalized posterior q), the ratio estimator of the previous subsection is used. It is slightly biased, with bias of order $1/S$, but consistent, and it is often more stable than the unnormalized-weight estimator [2, Sec. 3.3]. A common diagnostic for weight degeneracy is the effective sample size $S_{\text{eff}} = 1 / \sum_s (\tilde{w}^s)^2$; values much smaller than S signal that a few weights dominate.

4 Sampling-Importance Resampling / SIR (Section 10.4)

Sampling-Importance Resampling (also known as Importance Resampling) [1, Sec. 10.4] is used to draw *unweighted* approximate samples from the target distribution $p(\theta | y)$.

4.1 SIR Algorithm

To obtain k draws from $p(\theta | y)$:

Algorithm 2 Sampling-Importance Resampling (SIR)

- 1: Generate S candidate draws $\theta^1, \theta^2, \dots, \theta^S$ independently from proposal distribution $g(\theta)$ (where $S \gg k$).
- 2: Calculate raw weights $w(\theta^s) = \frac{q(\theta^s | y)}{g(\theta^s)}$ for $s = 1, \dots, S$.
- 3: Compute normalized probabilities:

$$p_s = \tilde{w}^s = \frac{w(\theta^s)}{\sum_{j=1}^S w(\theta^j)}$$

- 4: Resample k draws from candidate pool $\{\theta^1, \dots, \theta^S\}$ with replacement according to probabilities $(p_1, \dots, p_S)$.
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Remarks on resampling:

- Resampling *with* replacement, as in Algorithm 2 (the classical formulation of Rubin), is equivalent to multinomial resampling from the weighted pool. It is exact in the limit $S \rightarrow \infty$, since the empirical measure $\sum_s \tilde{w}^s \delta_{\theta^s}$ converges to the target, so the resampled draws are approximately distributed as $p(\theta | y)$ [2, Sec. 3.3].
- With replacement, draws with large weights appear several times. [1, Sec. 10.4] suggests sampling *without* replacement when k/S is small to reduce duplicates.
- Resampling adds Monte Carlo noise relative to using the weighted sample directly, so for estimating expectations the weighted estimator is preferable; resampling is useful when an unweighted sample is needed.

References

- [1] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd edition. Chapman & Hall/CRC, Boca Raton, FL.
- [2] Robert, C. P. and Casella, G. (2004). *Monte Carlo Statistical Methods*, 2nd edition. Springer, New York.